RADHIKA GOEL

Senior Software Engineer | Embedded Systems | USB Drivers | Automation

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Results-oriented Senior Software Engineer specializing in embedded systems, USB driver development, and automation, with a proven track record of accelerating validation, streamlining firmware delivery, and building robust, high-performance solutions across complex hardware-software environments.

AREAS OF EXPERTISE

C/C++ Programming | Python & Automation Scripting | USB Driver Development | RTOS & Embedded Systems | Bare-Metal | Automation & CI/CD Pipelines | Microservices Architecture | Linux System Development | Conversational AI & SDKs | Firmware Delivery & Validation

PROFESSIONAL EXPERIENCE

Synopsys India Private Limited, Noida/ Bangalore | Senior Software Engineer | Jul 2022 - Present

- Led automation of USB validation pipelines using Pytest and Jenkins, reducing manual effort by 50% and accelerating regression cycles.
- Developed a USB Test Environment Suite Automation (UTE), streamlining repetitive Git operations and enhancing version-control workflows for driver validation.
- Engineered bare-metal xHCI host and UDC device drivers, implementing register programming, TRB/DMA queue management, and interrupt handling for low-level USB communication.
- Streamlined firmware delivery by building containerized microservices using FastAPI, Docker, and MySQL.
- Optimized real-time device communication by integrating hardware-level bare-metal APIs across embedded systems.
- Implemented USB class drivers (CDC-ACM, MSC, and DFU) on MCU SDK with FreeRTOS, expanding USB functionality in RTOS-based environments.
- Expanded into the ATLAS AI team, developing FastAPI-based CRUD services for a Unified Installer Tool using Docker, Kubernetes, and MySQL.

INTERNSHIP

Intel Corporation, Bangalore | Graduate Technical Intern | Jul 2021 - Jun 2022

- Contributed to the development of a Conversational AI Chatbot SDK, supporting multiple AI-driven application use cases.
- Strengthened API reliability through systematic validation and automated testing using Robot Framework and Postman.
- Enabled scalable application deployment by containerizing and deploying microservices using Docker.
- Engineered real-time audio streaming capabilities using Streamlit, WebRTC, and VAD to support ASR inference workflows.
- Optimized speech-recognition model deployment by fine-tuning models and converting them to OpenVINO IR format.
- Improved code quality and reliability through debugging, logging, error handling, testing, peer reviews, and Git-based version control.

EDUCATION

M.Tech in Computer Science & Engineering, Artificial Intelligence - National Institute of Technology Hamirpur, Hamirpur

B.Tech in Computer Science & Engineering - APJ Abdul Kalam Technical University, Ghaziabad

ACADEMIC PROJECTS

- **Facial Recognition-based Image Captioning (M.Tech Thesis)**
  - Designed and implemented a **CNN + LSTM architecture** to generate contextual captions for images.
  - Integrated **OpenVINO Toolkit** for optimized face recognition and inference acceleration.
  - Achieved improved caption accuracy by fine-tuning pretrained deep learning models on custom datasets.
  - Demonstrated real-world applicability in **assistive technologies** and **surveillance systems**.
- **Image Classification using Deep Learning**
- **Movie Recommendation Engine**